

Calculus II Lesson 12

*n*th Term Test for Infinite Series $\sum_{n=1}^{\infty} a_n$

a) If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

b) If $\lim_{n \rightarrow \infty} a_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ might converge or might diverge.

Example: $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ and $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$ (series diverges)

$\lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n = 0$ and $\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \frac{1}{2}$ (series converges)

1)
$$\sum_{n=1}^{\infty} \frac{n^2}{n^2+1} = \frac{1^2}{1^2+1} + \frac{2^2}{2^2+1} + \frac{3^2}{3^2+1} + \cdots$$

a)
$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = \underline{\hspace{2cm}} ?$$

b) Use the n th Term Test to explain why the series $\sum_{n=1}^{\infty} \frac{n^2}{n^2+1}$ diverges/converges.

2) $\sum_{n=1}^{\infty} \frac{n+10}{10n+1}$

a) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n+10}{10n+1} = \underline{\hspace{2cm}} ?$ (Use L'Hopital's Rule)

b) Use the n th Term Test to explain why the series $\sum_{n=1}^{\infty} \frac{n+10}{10n+1}$ diverges/converges.

$$3) \sum_{n=1}^{\infty} \frac{2^n}{n^3}$$

$$\text{Hint: } D_x(a^x) = \ln(a) \cdot a^x$$

$$\text{a) } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{2^n}{n^3} = \underline{\quad ? \quad} \text{ (Use L'Hopital's Rule repeatedly.)}$$

$$\text{b) Use the } n\text{th Term Test to explain why the series } \sum_{n=1}^{\infty} \frac{n+10}{10n+1} \text{ diverges/converges.}$$

$$4) \sum_{n=1}^{\infty} \frac{2n^2 + 5n + 7}{4n^2 - 2n + 9}$$

$$a) \lim_{n \rightarrow \infty} \sum_{n=1}^{\infty} \frac{2n^2 + 5n + 7}{4n^2 - 2n + 9} = \underline{\quad ? \quad}$$

b) Use the n th Term Test to explain why the series $\sum_{n=1}^{\infty} \frac{2n^2 + 5n + 7}{4n^2 - 2n + 9}$ diverges or converges.

5) $\sum_{n=1}^{\infty} \frac{1}{n^3}$

a) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n^3} = \underline{\hspace{2cm}} ?$

b) What does the n th Term Test say about the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$?

6) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}}$

a) $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^3}} = \underline{\hspace{2cm}} ?$

b) What does the n th Term Test say about the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}}$?

7) $\sum_{n=1}^{\infty} \frac{1}{n}$

a) $\lim_{n \rightarrow \infty} \frac{1}{n} =$? _____

b) What does the n th Term Test say about the series $\sum_{n=1}^{\infty} \frac{1}{n}$?

8) Use the n th Term Test to explain why the following series diverges.

$$\sum_{n=1}^{\infty} \frac{n}{\ln n} \quad \text{Hint: Let } a_n = \frac{n}{\ln n}$$

a) Find $\lim_{n \rightarrow \infty} a_n =$ _____

b) Explain why the series $\sum_{n=1}^{\infty} \frac{n}{\ln n}$ diverges.

9) Use the n th Term Test to explain why the following series diverges.

$$\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right) \quad \text{Hint: Let } a_n = \ln\left(\frac{n+1}{n}\right)$$

a) Find $\lim_{n \rightarrow \infty} a_n =$ _____

b) Explain why the series $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$ diverges.

10) Use the n th Term Test to explain why the following series diverges.

$$\sum_{n=1}^{\infty} \cos n \quad \text{Hint: Let } a_n = \cos n$$

a) Find $\lim_{n \rightarrow \infty} a_n =$ _____

b) Explain why the series $\sum_{n=1}^{\infty} \cos n$ diverges.

